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17 June 1975

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MILITARY STANDARD

# TAPE, VIDEO, MAGNETIC, RECORDING, FORMATS FOR

FSC 5820

HEADQUARTERS
DEPARTMENT OF DEFENSE
WASHINGTON, DC 20301

Tape , Video, Magnetic, Recording,
Formats for

MIL-STD-1856A

1. This Military Standard is approved for use by all Departments and Agencies of the Department of Defense.

2. Recommended corrections, additions or deletions should be addressed to Commander, US Army Electronics Command, ATTN: AMSEL-RD-TS Fort Monmouth, NJ 07703.



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# FOREWORD

This is the first book format standard generated on magnetic video tape recording formats. This standard is approved for use by all departments and agencies of the Department of Defense. It is intended to prevent the entry of unnecessary items (sizes, types, varieties) into the Department of Defense Logistics System. This is not a procurement document. This document is not intended to restrict any activity in selecting new items required to support state-of-the-art changes.



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# TAPE, VIDEO, MAGNETIC, RECORDING, FORMATS FOR

## 1. SCOPE

1.1 Purpose. This standard establishes recording formats for magnetic video recording to achieve the highest practical degree of compatibility among equipments of the same type.

1.2 Classification.

1.2.1 Types. Video magnetic tape recording formats covered by this standard shall be classified as follows:

| | |
|---|---|
| TYPE I | 2-inch Quadruplex, Reel-to-Reel, Format for. |
| TYPE II | 1-inch Helical Scan, Reel-to-Reel, Format for. |
| TYPE III | 3/4-inch Helical Scan, Cassette, Format for. |
| TYPE IV | 1/2-inch Helical Scan, Reel-to-Reel, Format for. |

## 2. REFERENCED DOCUMENTS

2.1 The documents referenced below form part of this standard to the extent specified herein. Unless otherwise specified, the issue in effect on the date of invitation for bids or request for proposals shall apply.

American National Standards Institute

| | |
|---|---|
| C98.1-1963 (R1969) | 2-inch Video Magnetic Tape, Dimensions of. |
| C98.4-1970 | Speed of 2-inch Tape for Quadruplex Video Magnetic Tape Recording. |
| C98.5-1970 | 2-inch Video Magnetic Tape Reels, Dimensions of. |
| C98.6-1973 | Video, Audio and Tracking Control Records on 2-inch Video Magnetic Tape, Dimensions of. |

(Application for copies should be addressed to the American National Standards Institute, Inc., 1430 Broadway, New York, NY 10018)



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3. DEFINITIONS - None

4. GENERAL REQUIREMENTS

TYPE I - This format conforms to the American National Standards Institute standards in paragraph 2.

TYPE II - This format conforms to Figure 1 of this document.

TYPE III - This format conforms to Figure 2 and Table I of this document. The cassette conforms to Figures 3, 4, 5, 6, and 7 of this document.

TYPE IV - This format conforms to Figure 8 and Table II of this document.

5. DETAILED REQUIREMENTS - None other than those listed in Section 4.

Custodians:

Army - EL
Navy - EC
Air Force - 80

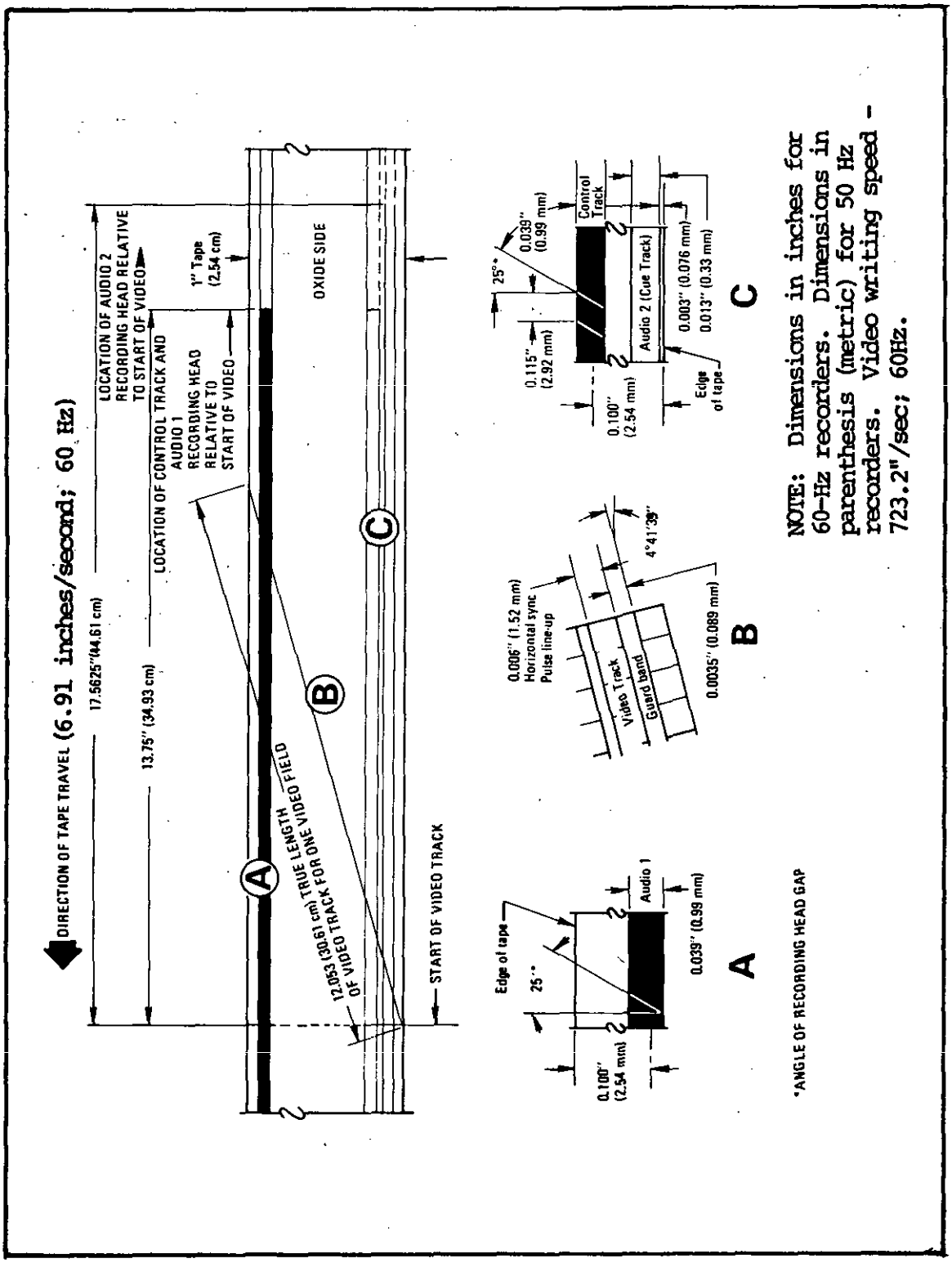
Preparing Activity:

Army - EL

Project No. 5820-0054



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NOTE: Dimensions in inches for 60-Hz recorders. Dimensions in parenthesis (metric) for 50 Hz recorders. Video writing speed - 723.2"/sec; 60Hz.

Figure 1. ONE-INCH, HELICAL SCAN, REEL-TO-REEL, FORMAT FOR.



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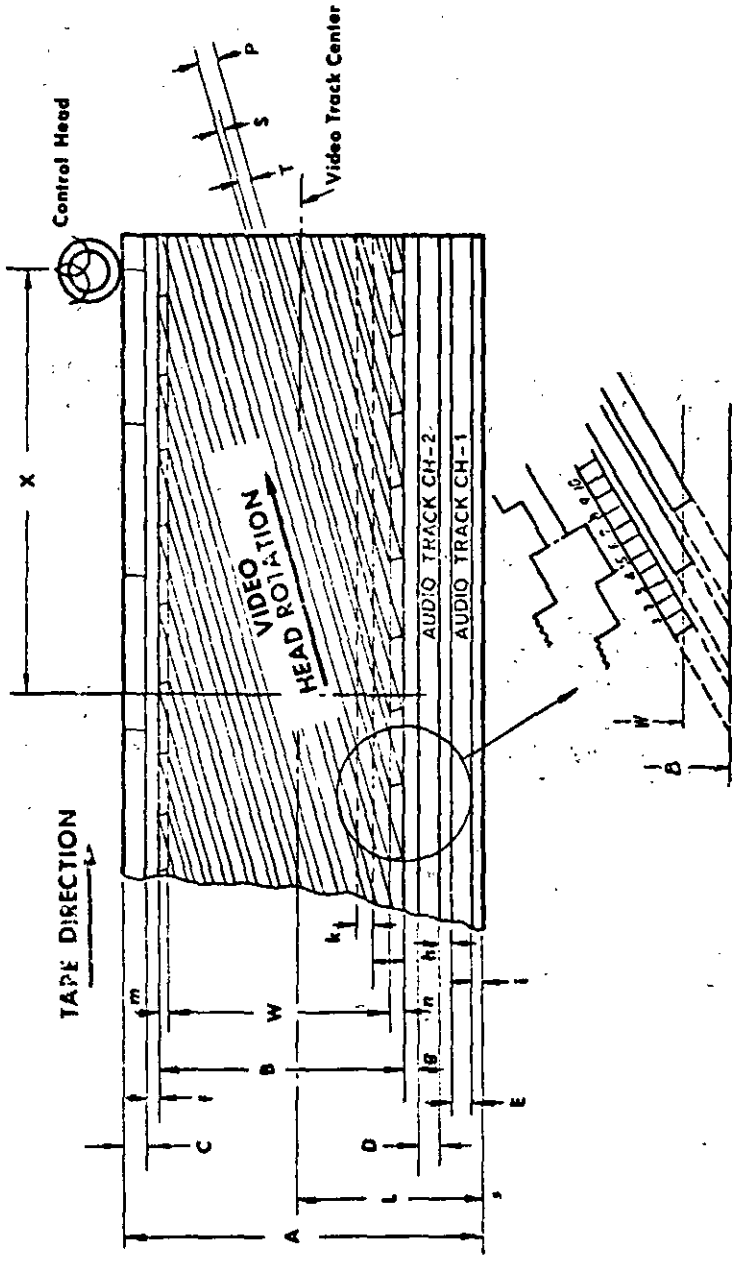



Figure 2. THREE-FOURTHS-INCH HELICAL SCAN, CASSETTE, FORMAT FOR.



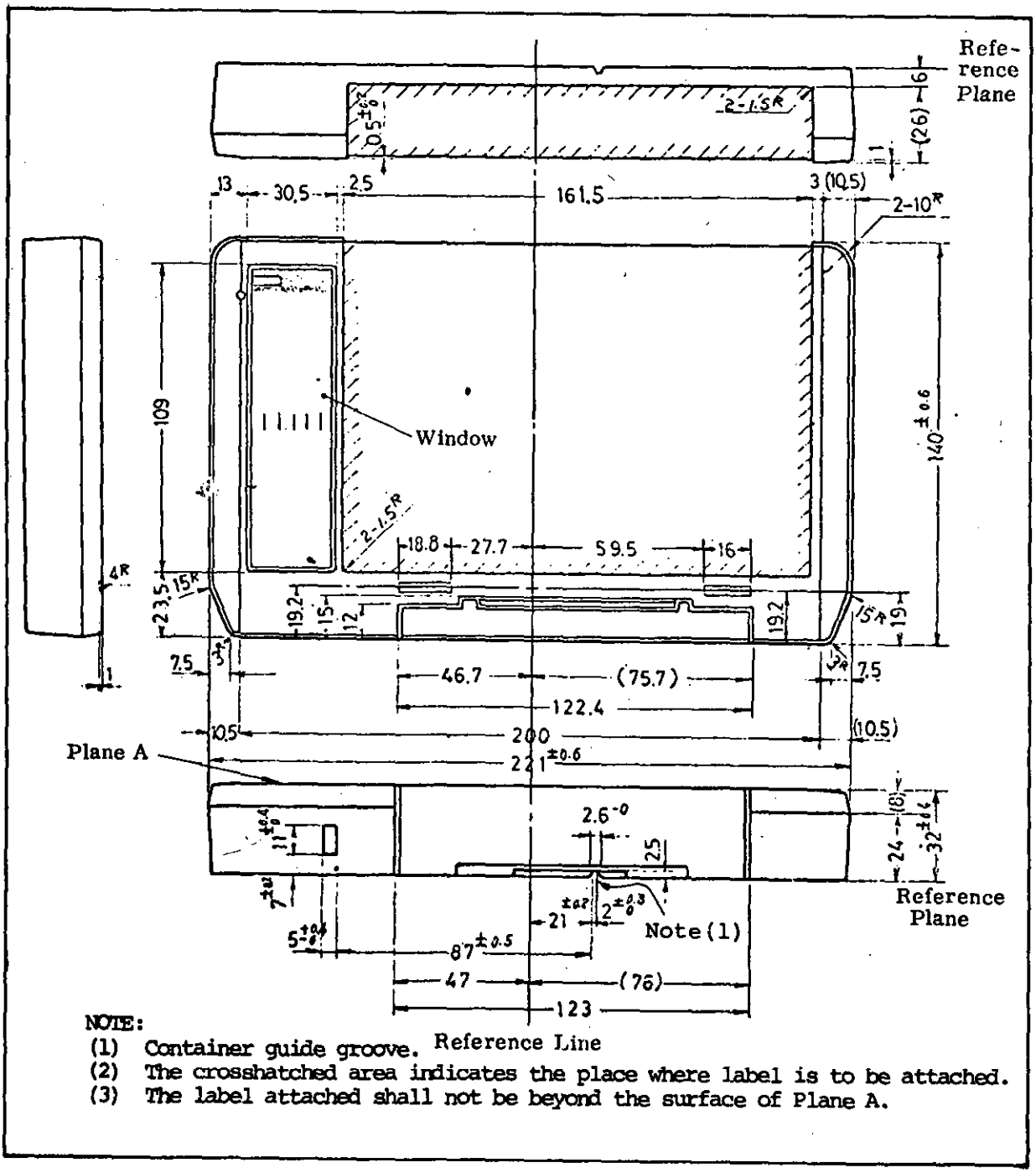
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TABLE I

| Symbol | Item | |
|---|---|---|
| A | TAPE WIDTH | 19.00 mm ± 0.03 |
| Vt | TAPE SPEED | 95.3 mm/s ± 0.2% |
| Vr | VIDEO WRITING SPEED | 10.26 m/s |
| P | VIDEO TRACK PITCH | 0.1374 mm |
| B | VIDEO RECORDING ZONE | 15.5 mm |
| W | VIDEO RECORDING ZONE EFFECTIVE WIDTH | 14.8 mm |
| L | CENTER OF VIDEO TRACK | 10.450 ± 0.05 |
| T | VIDEO TRACK WIDTH | 0.085 ± 0.007 |
| C | CONTROL TRACK WIDTH | 0.6 mm |
| D | AUDIO CH-2 TRACK WIDTH | 0.8 ± 0.05 mm |
| E | AUDIO CH-1 TRACK WIDTH | 0.8 ± 0.05 mm |
| S | VIDEO - VIDEO GUARD BAND | 0.052 mm |
| f | CONTROL - VIDEO GUARD BAND | 0.2 mm |
| g | AUDIO - VIDEO GUARD BAND | 0.2 mm |
| h | AUDIO - AUDIO GUARD BAND | 0.7 mm |
| i | EDGE - AUDIO GUARD BAND | 0.2 ± 0.1mm |
| m | BACK OVERLAP | 0.35 mm |
| n | FRONT OVERLAP | 0.35 mm |
| o | VIDEO TRACK ANGLE | |
| | TAPE STATIONARY | 4° 54' 49.1" |
| | TAPE MOVING | 4° 57' 33.2" |
| | VIDEO TRACK LENGTH | 171.2 mm |
| k | CUE TRACK WIDTH | 0.5 ± 0.05 mm |
| q | CUE TRACK TO LOWER LIMIT OF VIDEO TRACKS | 1.1 ± 0.05 mm |
| X | DISTANCE FROM THE END OF 180° SCAN OF VIDEO HEAD TO AUDIO AND CONTROL HEAD POSITION ON THE TAPE | 74 mm |



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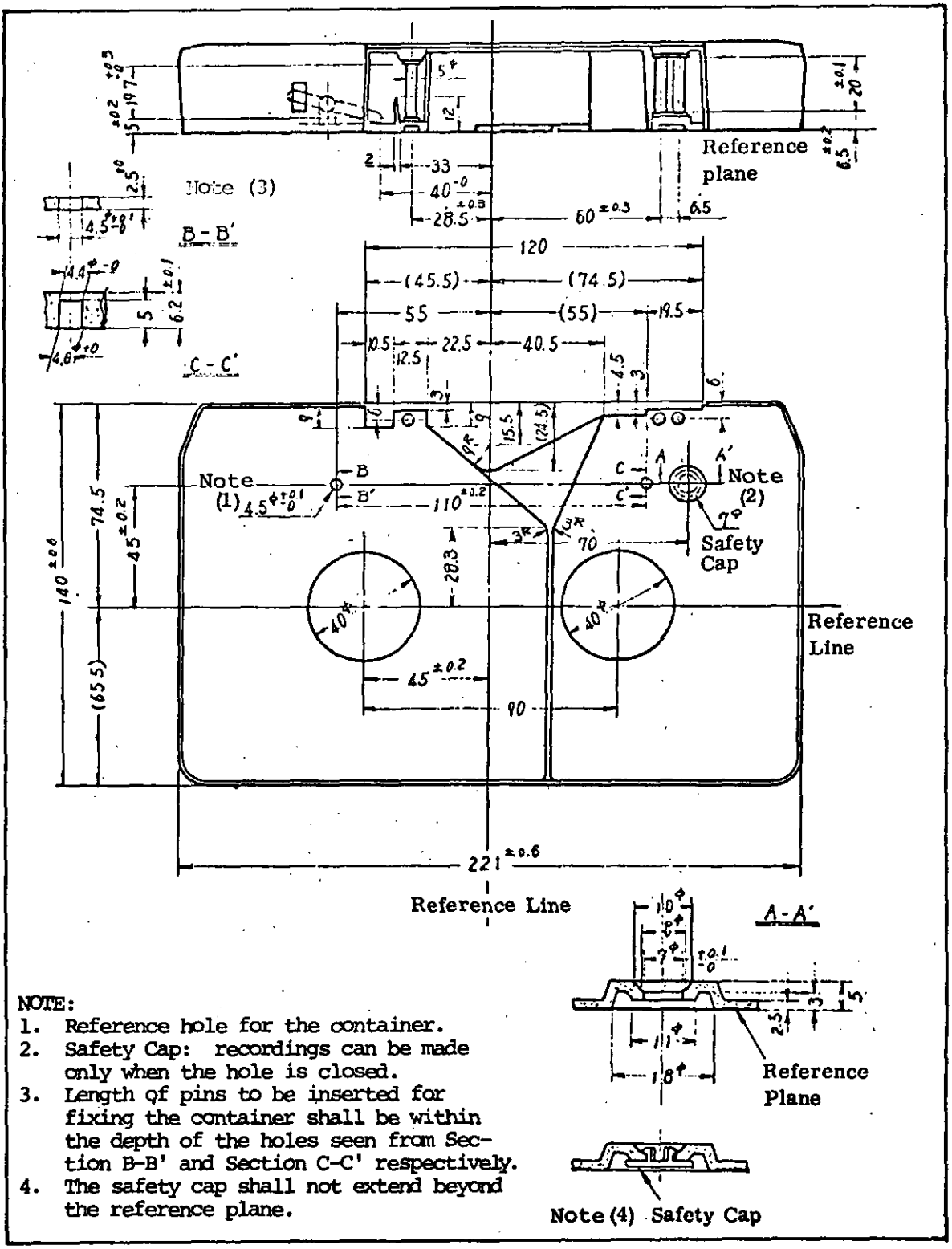



NOTE:
(1) Container guide groove.
(2) The crosshatched area indicates the place where label is to be attached.
(3) The label attached shall not be beyond the surface of Plane A.

Figure 3. THREE-FOURTHS-INCH MAGNETIC TAPE CASSETTE.



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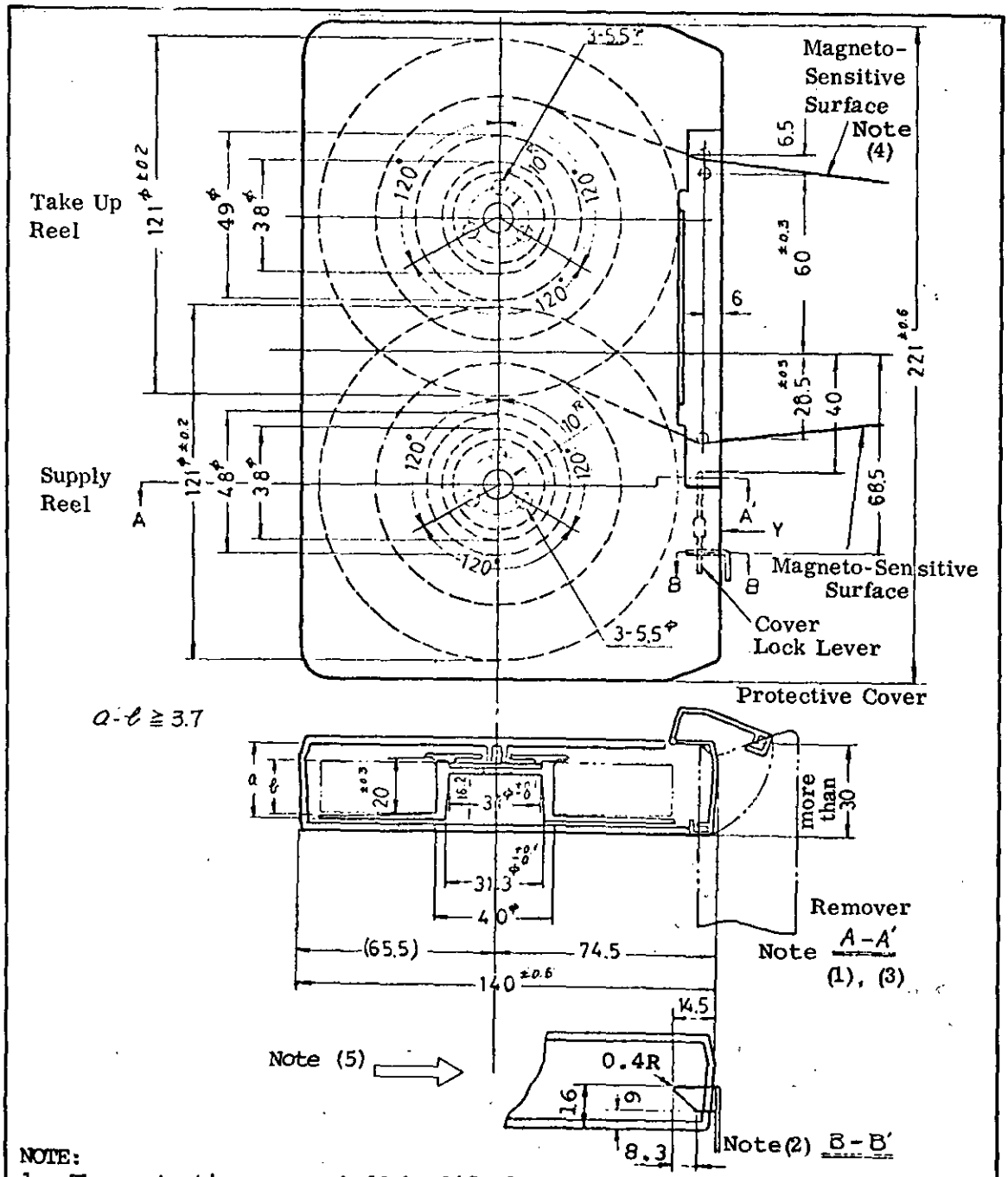



NOTE:
1. Reference hole for the container.
2. Safety Cap: recordings can be made only when the hole is closed.
3. Length of pins to be inserted for fixing the container shall be within the depth of the holes seen from Section B-B' and Section C-C' respectively.
4. The safety cap shall not extend beyond the reference plane.

Figure 4. THREE-FOURTHS-INCH MAGNETIC TAPE CASSETTE.



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NOTE:
1. The protective cover shall be lifted more than 30 mm in Section A-A'.
2. Indicates dimensions when the cover lock lever released by latch seen from Section B-B' is inserted into the container.
3. Indicates the position of the protective cover remover when the container seen from Section A-A' is placed on the reels.
4. The magneto-sensitive surface of the tape shall be outside.
5. The arrow shows the direction of the container to be inserted.

Figure 5. THREE-FOURTHS-INCH MAGNETIC TAPE CASSETTE.



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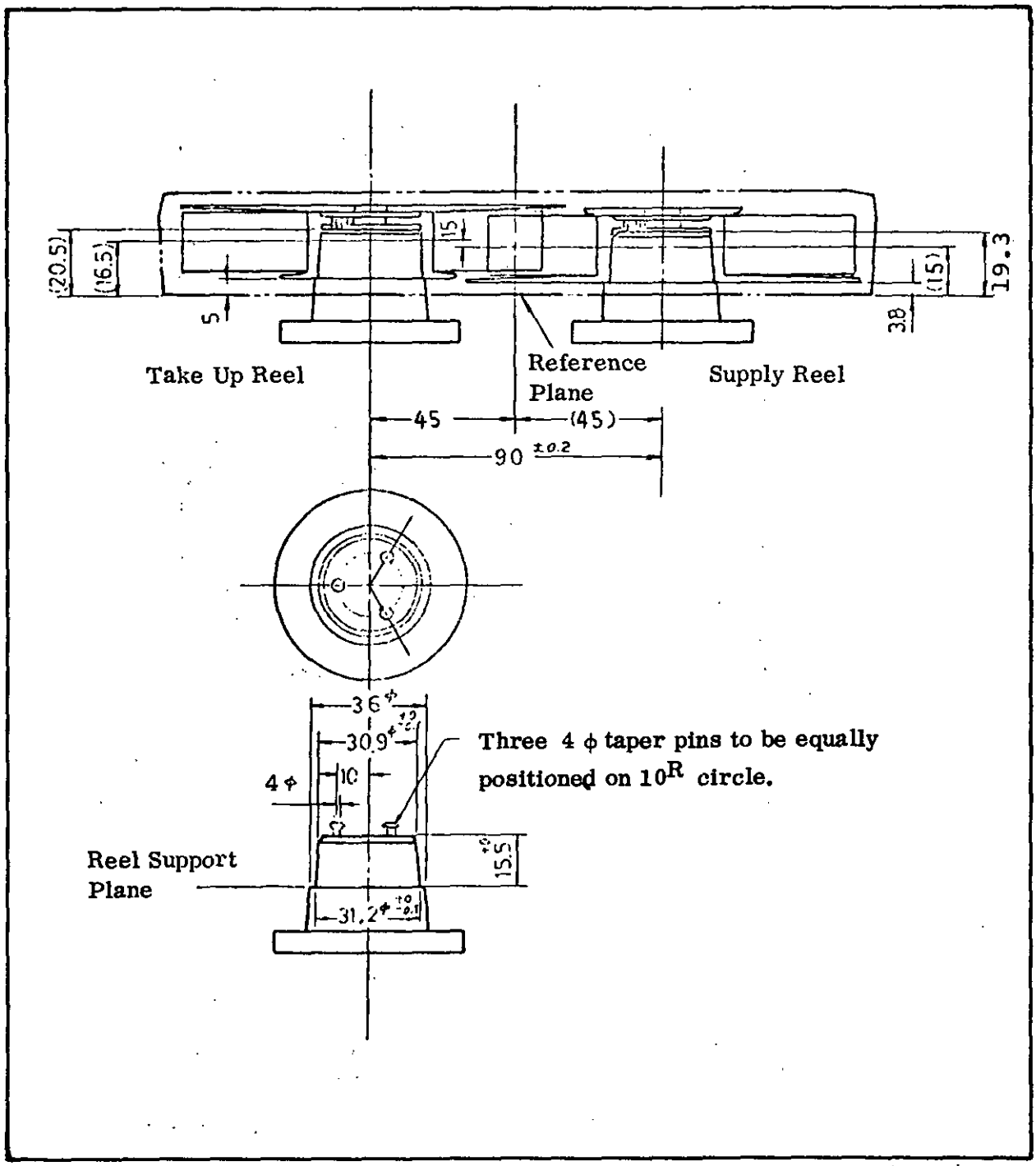
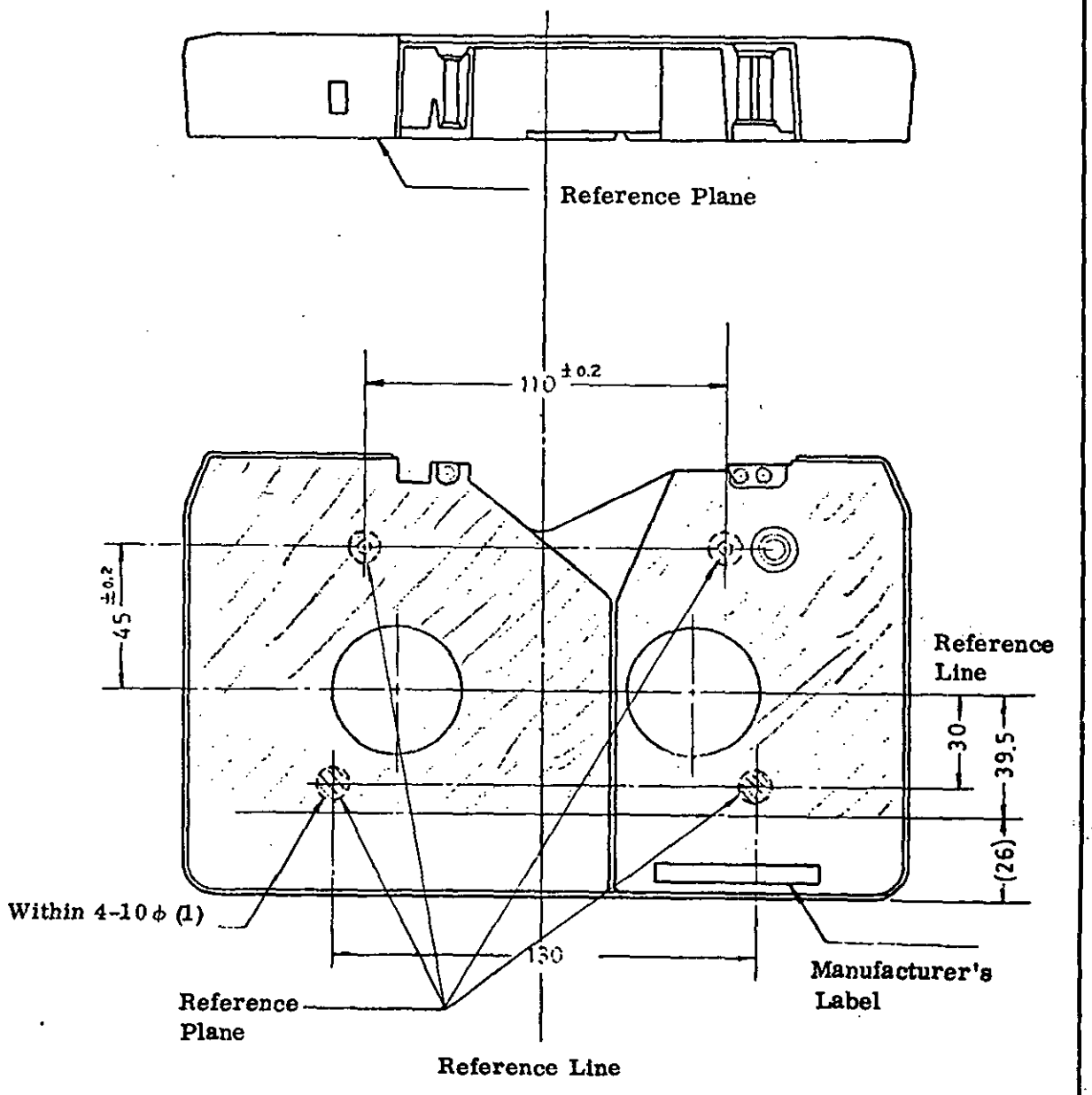



Figure 6. THREE-FOURTHS-INCH MAGNETIC TAPE CASSETTE.



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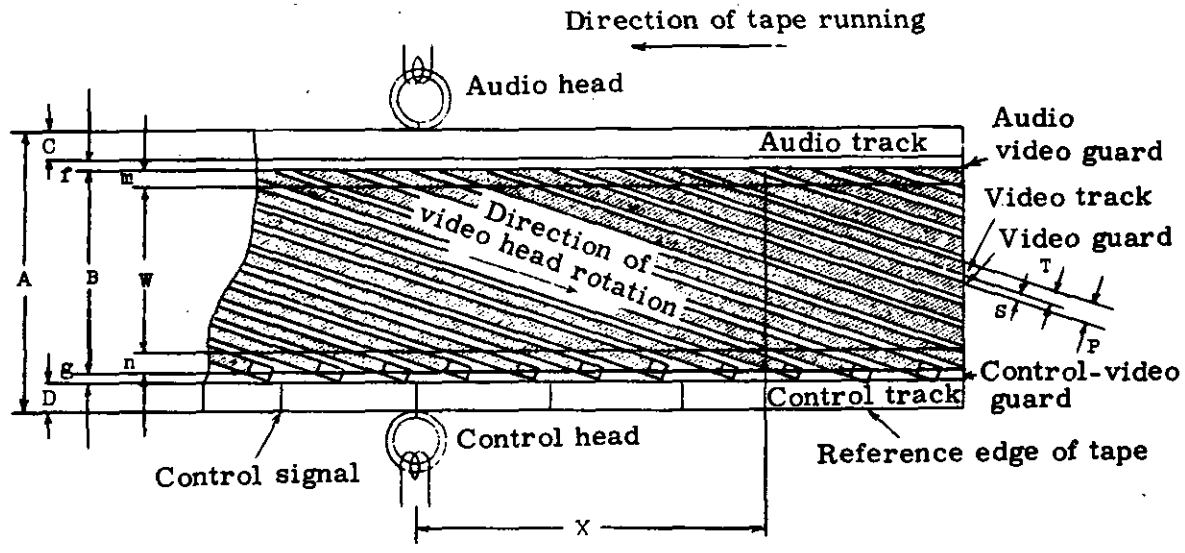

NOTE: The flatness of the crosshatched portion of the reference plane shall be within 0.2 mm.

Figure 7. THREE-FOURTHS-INCH MAGNETIC TAPE CASSETTE.



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MAGNETIC TAPE FORMAT




NOTE: The figure shows the magnetic coated surface of the tape.

| Symbol | (in mm) |
|---|---|
| A | $12.7^{+0.0}_{-0.1}$ |
| B | 10.65 |
| C | $1.0^{+0.0}_{-0.1}$ |
| f | 0.15 |
| m | 0.275 |
| w | 10.10 |
| n | 0.275 |
| g | 0.1 |
| D | 0.8 |
| P | 0.173 |
| S | more than 0.04 |
| T | more than 0.1 |
| X | 81.0 |

REMARKS:
(1) The video track can be recorded in the control-video guard band.
(2) It is allowable if a part of the "scan beginning overlap width (m)" or a part of the "scan ending overlap width (n)" is missing from the video track. But the range of the video width (W) shall be correct.

Figure 8. ONE-HALF-INCH, HELICAL SCAN, REEL-TO-REEL, FORMAT FOR.



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TABLE II

| Symbol | Item | | |
|---|---|---|---|
| A | TAPE WIDTH | mm | $12.7\ ^{+0}_{-0.1}$ |
| Vt | TAPE SPEED | mm/sec | 190.5 ±0.5% |
| $\phi$ | DRUM DIAMETER | mm | 115.82 ±0.01 |
| Vh | WRITING SPEED | m/sec | 11.1 |
| P | VIDEO TRACK PITCH | mm | 0.173 |
| B | FULL VIDEO WIDTH | mm | 10.65 |
| W | VIDEO WIDTH | mm | 10.10 |
| C | AUDIO TRACK WIDTH | mm | 1.0 |
| D | CONTROL TRACK WIDTH | mm | 0.8 |
| f | AUDIO-VIDEO GUARD WIDTH | mm | 0.15 |
| g | CONTROL-VIDEO GUARD WIDTH | mm | 0.1 |
| m | SCAN BEGINNING OVERLAP WIDTH | mm | 0.275 |
| n | SCAN ENDING OVERLAP WIDTH | mm | 0.275 |
| $\theta_0$ | VIDEO TRACK ANGLE (STOPPING) | | $3^{\circ}11'$ |
| $\theta$ | VIDEO TRACK ANGLE (RUNNING) | | $3^{\circ}7'43"$ |
| $\ell$ | VIDEO TRACK LENGTH | mm | 185.1 |
| X | POSITIONS OF AUDIO AND CONTROL HEADS | mm | 81.0 |

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# SPECIFICATION ANALYSIS SHEET

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INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION MIL-STD-1856A TAPE, VIDEIO, MAGNETIC, RECORDING, FORMATS FOR

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

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